



New Engineering Onboarding

Why this project is noteworthy.

Onboarding is one of the most important integration points for a new hire and a business. It can no longer be just a torrent of perfunctory exercises, “here’s your stuff, sit there and go to work” approach.

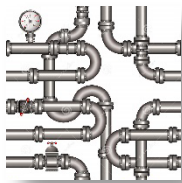


Research proves that companies with successful onboarding programs are not just more likely to retain their new hires but report measurable profit growth. Those that excel in leadership development, talent management, and performance management experience substantially higher revenue growth and profit margins. Onboarding is where this mutually successful relationship begins. The new employee needs to be introduced to the culture, the legacy, and be invited into the future opportunity that lies before them.

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Overview

A major California utility had a significant challenge... tens of thousands of miles of aging gas transmission and distribution pipelines were in continuous need of maintenance, repair, and expansion. Analog solutions were expensive and slow. New ideas and advanced digital solutions needed to be engineered to ensure the safe delivery of an inherently dangerous form of energy while remaining profitable. To address this issue, top students from leading engineering schools were targeted and recruited for key engineering positions. Training and retaining this top talent was a priority for the Gas Operations’ team.



Business Goals

Gas Operations goal: Create a pipeline of highly skilled gas engineers equipped to understand and meet future Gas Operations’ challenges and needs.

Their specific needs were to:

- create measurable programs for attracting, training, and retaining top engineering talent
- reduce:
 - Costs associated with key personnel turn-over
 - New engineering time to proficiency
 - Program redundancy
- address the unique challenges new engineer students face in a corporate environment



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Analysis

To determine the optimal leverage points that would bring about the highest degree of impact, interviews and focus groups were conducted with the following stakeholders: (listed are high-level findings)

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- Senior engineers (managers and supervisors of new engineers):
 - Candidates were lacking specific job area expertise
 - Candidates had great ideas, but needed to uplevel their ability to:
 - work in teams (collaborate)
 - communicate
 - present out their solutions professionally
- Program graduates focus groups:
 - Class materials were too extensive and detailed
 - Sequence of content was arbitrary and presenter schedule driven
 - Overlap of content between onboarding programs (company orientation, gas orientation, engineering orientation)
 - Presentations were long and lecture only
 - Did not have clear understanding of all available career opportunities
 - No connection to other new hires
 - Locating specific class information while on the job was difficult
 - Integration into the culture was lacking

Additional research was conducted to understand and address general onboarding program

opportunities and inherent issues. A best practices for talent retention of millennials (target hiring group) was conducted through reviewing published Gallup, Harvard Business Review, and The Aberdeen Group research.

What Millennials Want		
...from their boss	...from their company	...to learn
TOP FIVE CHARACTERISTICS MILLENNIALS WANT IN A BOSS	TOP FIVE CHARACTERISTICS MILLENNIALS WANT IN A COMPANY	TOP FIVE THINGS MILLENNIALS WANT TO LEARN
Will help me navigate my career path	Will develop my skills for the future	Technical skills in my area of expertise
Will give me straight feedback	Has strong values	Self-management and personal productivity
Will mentor and coach me	Offers customizable options in my benefits/reward package	Leadership
Will sponsor me for formal development programs	Allows me to blend work with the rest of my life	Industry or functional knowledge
Is comfortable with flexible schedules	Offers a clear career path	Creativity and innovation strategies

The current onboarding program, vendor training, and curriculum approach was reviewed. There were significant redundancies and very little accountability for learning the materials across all programs. Also missing was any connection to the rich engineering culture the utility had developed during its 100+ years of existence.



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Learning Goals

CA Consulting collaborated with senior Gas Operations' personnel to further frame the issues surfaced during research and to design and develop a program strategy that addressed critical business needs. Our new initiative needed to be tightly integrated into existing recruiting and vendor training programs.

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A significant challenge existed with one vendor developed course:

- The materials were very dense
- There weren't any assessments
- Some content from the vendor course overlapped with our content
- Not all participants in the onboarding class would have to take this course
- The vendor wanted too much money to update the course

Our solution was to have participants in the vendor course responsible for teaching back only the redundant content elements to participants in the onboarding class. This increased their initial accountability for the content in the vendor class, gave them a chance to work on their presentation skills, and removed the redundancy of their having to sit through the content again.

Solution:

- Two weeks of content was architected into a logical curriculum map tracking how a project flowed from ideation, to budget, to design, to build/maintenance
- Content scope was reduced to "What a new engineer needs to know today"
- To facilitate learning, discussion topics and questions were added for each of the segments
- Lectures were broken-up with:
 - Field trips to strategic engineering locations
 - Short video profiles from recent graduates
 - Virtual Q & A with senior engineers and recent program grads
 - Pair and group discussions and presentations, role-playing, and problem-solving activities
- Communications, collaboration, and presentation training were integrated into the curriculum
 - Participants practiced new skills in the class through pair and share exercises, teach-back opportunities, and final project presentations
 - Skills were peer reviewed and feedback was provided
- A timeline-based video presentation, "New Pioneers", highlighting the rich engineering history, was written, and produced. The video illustrated how innovative engineering solutions had helped solve many of the company's challenges over the years. The final minute was a direct invitation for each participant to step into the opportunity and to add to the rich engineering legacy by becoming "a new pioneer" of the utility. The video was followed up by a live presentation from leadership.



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Assessment

To determine how well the content being presented was being retained, a team-based, progressive project was instituted:

- Teams worked on an actual project throughout the two-week course
- Each team analyzed and completed tasks related to different aspects of Gas Operations
- Teams worked collaboratively to solve the problems, develop the presentations, and present out their findings to the class
- All class members provided final feedback on the communications and presentations

- The entire two-week curriculum was migrated from a paper-based delivery format to a searchable e-book. This allowed participants to search and reference class materials on the job.
- A manager and field mentoring training was developed to support new hires in the field
- A cloud-based app was incorporated for connecting the cohort, managers, and mentors

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RESULTS

Results:

The project roll-out is in its initial phases. The initial level one feedback from participants and instructors was extremely positive. The Progressive Project reports demonstrated a much greater command of the

information taught. The comparative year to year scores for level three Kilpatrick focus group feedback showed a significant rise in application of the materials in the field.



Strategy



Program
Development



Instructional
Design



Writing